



**Verified Carbon  
Standard**

## IMPROVED COOKSTOVES DISTRIBUTION IN SIERRA LEONE - SOUTHERN REGION

|                                  |                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project title</b>             | <i>Improved cookstoves distribution in Sierra Leone - southern region</i>                                                                                                                                                                                                                                                                    |
| <b>Project ID</b>                | 5320                                                                                                                                                                                                                                                                                                                                         |
| <b>Crediting period</b>          | <i>1<sup>st</sup> June 2025 – 31<sup>st</sup> May 2035</i>                                                                                                                                                                                                                                                                                   |
| <b>Original date of issue</b>    | <i>28<sup>th</sup> October 2024</i>                                                                                                                                                                                                                                                                                                          |
| <b>Most recent date of issue</b> | <i>28<sup>th</sup> October 2024</i>                                                                                                                                                                                                                                                                                                          |
| <b>Version</b>                   | 1                                                                                                                                                                                                                                                                                                                                            |
| <b>VCS Standard Version</b>      | 4.7                                                                                                                                                                                                                                                                                                                                          |
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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

The clean cooking project developed by Eni S.p.A. is a voluntary Grouped Project (“GP”) involving the distribution of Improved Cookstoves (“ICSs”) among local communities in the rural areas of the districts of Pujehun, Kenema and Kaliahun, in the Southern Region of Sierra Leone (“Country”), where the majority of households still rely on inefficient and pollutant traditional cooking techniques.

The activities in field are performed by Eni together with a Project Instance (“PI”) field partner called CUAMM (Doctors with Africa) (“CUAMM”). CUAMM has been developing sustainable projects and training/educational activities in the Country since 2012 with Health-related activities and projects. CUAMM has two different offices in the country (Freetown & Pujehun), creating a deep and strong relationship with local communities. CUAMM oversees communities’ engagement, ICSs distribution (storage, management, last-mile distribution and households training) and follow up with recipient beneficiary households for assisting them in the use of the ICS.

The project aims to replace the traditional cooking systems currently used by local people (i.e. three-stone fires or rudimental braziers) with ICSs that require less fuel and thus decrease the biomass combustion’s emissions, with a consequent reduction of health hazards coming from smoke pollution. The switch to more efficient stoves will help limiting the demand of woody biomass, reducing the unsustainable forests cuttings and contrasting deforestation which is progressing rapidly in the Country. The improved cookstoves’ deployment will also have a positive impact on women’ and children’ life, reducing the time they spend collecting wood and cooking meals, allowing them to invest their time in other activities.

The ICS beneficiaries voluntarily participate in the GP, signing an End User Agreement (“EUA”) with Eni confirming that any emission reductions resulting from the use of the ICS delivered is owned by Eni. The personal information of the beneficiary household and the serial number of the ICS are noted in the EUA and stored in a data management system which allows Eni to uniquely identify each ICS by tracking the GPS coordinates of the beneficiary household and the ICS delivery date.

According to the project framework, ICSs will be distributed to those households that prior to project implementation, as confirmed during baseline data collection, were relying on the use of three-stone fires or traditional braziers with no improved combustion air supply or chimney, for their cooking activities. Each household targeted by the GP will receive an ICS according to wood biomass used as cooking fuel.

The current baseline scenario is characterized by the usage of inefficient cooking techniques with poor ventilation and high heat loss. To achieve the same thermal energy, according to the

Clean Cooking Alliance<sup>1</sup>, traditional cooking methods could require more than twice the amount of wood fuel compared to the ICS's utilization.

The prosecution of the baseline scenario would see:

- Increasing woody biomass consumption with consequent aggravated deforestation and/or forest degradation and loss of biodiversity in the Country;
- Increasing distances and time to be walked by women and children to collect wood for cooking;
- Increasing incidence of health risks for women and children who inhale the smoke and particulate matter produced by the combustion with traditional systems.

This project is developed in the southern region of Sierra Leone and the implementation areas are the districts of Pujehun, Kenema and Kaliahun in rural areas.

The average annual GHG emission reduction generated by the implementation of the project in the southern region is around 164.114 tCO<sub>2</sub>e per year (almost 1,641 MtCO<sub>2</sub>e in the entire project's lifetime), with lower emission reduction in the initial and last years of the project and a peak in the middle of the project's lifetime due to the ICSs distribution profile.

## 1.2 Audit History

| Audit type        | Period | Program | Validation/verification body name | Number of years |
|-------------------|--------|---------|-----------------------------------|-----------------|
| <i>Validation</i> | NA     | VCS     | NA                                | <i>One year</i> |

## 1.3 Sectoral Scope and Project Type

|                              |                                               |
|------------------------------|-----------------------------------------------|
| <b>Sectoral scope</b>        | 03 - Energy Demand                            |
| <b>Project activity type</b> | Type II - Energy Efficiency Improved Project; |

## 1.4 Project Eligibility

### 1.4.1 General eligibility

<sup>1</sup> <https://cleancooking.org/wp-content/uploads/2023/08/Climate-Environment-and-Clean-Cooking.pdf>

The project activity is included under the scope of the VCS Program in the category of efficiency improvements in thermal applications. Indeed, the project activity is not mentioned in the list of activities excluded from the VCS Program Scope (Table 1, VCS Standard v4.7)

The methodology applied to the project (Energy efficiency and fuel switch measures in thermal applications, VM0050 v1.0) is eligible under the VCS Program because it has been developed and approved by Verra.

The VCS Program also excludes the following project activities under the circumstances indicated in Table 1, below, as indicated in the VCS standard version 4.7.

*Table 1: Excluded Project Activities*

| Exclusions from VCS Program Scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Exclusion                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity), whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition. | Excluded in non-LDCs (Low Developed Countries). Further, large-scale projects excluded in LDCs.<br><br>Since the PI is meant to deliver improved cookstove to local communities, the technology does not involve any grid-connected electricity generation. |
| Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.                                                                                                                                                                                                                                      | Excluded in Non-LDCs<br><br>As mentioned above, the PI is meant to distribute improved cookstove, fueled by wood, and does not generate grid – connected electricity                                                                                        |
| Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).                                                                                                                                                                                                                                                                                                                                         | Excluded in Non-LDCs<br><br>The PI does not involve heat recovery for combined cycle electricity generation, cogeneration and trigeneration activities                                                                                                      |
| Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Excluded in Non-LDCs                                                                                                                                                                                                                                        |

|                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO2 capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves)</p>                                     | <p>The end user of the technology provided by PP (Project Proponent) are the households. Moreover, the stoves are designed for domestic use, not industrial one, so the project and the technology do not generate electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.</p> |
| <p>Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).</p>                              | <p>Excluded for large-scale projects in non-LDCs.</p> <p>The PI does not involve any fuel switch nor the use of fossil fuel.</p>                                                                                                                                                                                                                                           |
| <p>Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO2e per year.</p> | <p>Excluded for large-scale projects in non-LDCs.</p> <p>The PI does not involve any activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).</p>                                                             |
| <p>Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO2e per year</p>                                                                                                            | <p>Excluded for large-scale projects in non-LDCs.</p> <p>The PI does not provide activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.</p>                                                                                                                                                                           |
| <p>Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions</p>                                                                                                                                                                                                                                                                                                                | <p>Excluded in all countries.</p> <p>The PI does not involve activities that reduce HFC-23 emission.</p>                                                                                                                                                                                                                                                                   |

### 1.4.2 AFOLU project eligibility

This section is not applicable because the GP concerns with the energy efficiency improvement of thermal applications adopted by local communities for cooking.

The GP does not belong to the AFOLU projects category.

### 1.4.3 Transfer project eligibility

This section is not applicable because the GP has been listed solely in the VCS Program and it is seeking registration only in Verra registry.

The GP has not been listed in other carbon registries.

## 1.5 Project Design

The project has been designed as “Grouped project”.

### 1.5.1 Grouped project design

According to the framework of Grouped Projects, each new Project activity Instance is going to meet the eligibility criteria required by VCS Standard v4.7 and the methodology applied (VM0050 V1.0). The eligible criteria are summarized in the following Table .

Table 2: Set of eligibility criteria for new Project activity instances

| No. | Criterion                                                                           | Explanation of how the new Project activity Instances match the criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Meet the applicability conditions set out in the methodology applied to the project | <p>The ICSs distributed in a new PI will meet the applicability conditions set out in Section 4 of VM0050 v1.0 methodology:</p> <ul style="list-style-type: none"> <li>• ICSs will replace non-renewable biomass-fired cookstoves with more efficient project devices that use the same fuel as in the baseline.</li> <li>• ICSs will be issued to households.</li> <li>• Project devices are single-pot portable stoves with at least 25% thermal efficiency and powered by woody biomass (wood).</li> <li>• The incentive mechanisms are implemented by the project partner and includes trainings sessions for lead mothers. The training covers financial management, the use and maintenance of improved cookstoves, and lessons on health, nutrition, and environmental awareness. This approach empowers lead mothers to educate their communities on these topics that lead to behavioural changes.</li> </ul> |

|   |                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                                                                                                                    | <ul style="list-style-type: none"> <li>When the project device will end its technical life (5years), no further emission reductions will be claimed for the project device.</li> <li>The double counting of emissions reduction will be avoided collecting the ICSs' serial number and the GPS coordinates of each beneficiary household.</li> <li>There is no specific regulatory framework in place regarding cookstoves distribution, commercialization, installation and use of improved cookstoves and fuel supply and use for the target population.</li> <li>There is currently only one Carbon Offset REDD project underway in the Country: Gola REDD+. This project is located in the southeastern districts of Kailahun, Pujehun and Kenema, however the project boundary of the Gola REDD+ does not overlaps with the expected fuel source area of the project. The ICSs will be distributed in different areas, since the Gola REDD+ cover remote and not populated areas.<sup>2</sup></li> </ul> <p>An agreement is signed between the households involved and Eni: each household receives the ICS without cash uptake in exchange for Eni ownership of the environmental benefits' property resulting from the ICS use.</p> |
| 2 | Use the technologies or measures specified in the project description.                                                             | The ICSs distributed for new Project Instances will have at least the same technical features of the ones delivered at the start of the project, and in any case thermal efficiency higher than 25% as set out in VM0050 v1.0 methodology.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3 | Apply the technologies or measures in the same manner as specified in the project description.                                     | The ICSs will substitute the inefficient cooking techniques that are still adopted by the local communities. During the ICS delivery, information regarding biomass use as fuel and cooking baseline appliances will be collected and stored in a data management system in order to facilitate the following validation and verification processes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4 | Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area. | The new Project activity Instances will be issued in the geographical areas identified in Section 1.13, therefore the baseline scenario is the same described in Section <b>Errore</b> . <b>L'origine riferimento non è stata trovata.</b> (traditional inefficient cooking methods with wood).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

<sup>2</sup>

[https://static1.squarespace.com/static/5d2c7fcb94060e0001e56248/t/627cd5c8eab77d1e1693fab1/1652348403875/VCS\\_Project+Description\\_1201\\_10Jan2021\\_submitted.pdf](https://static1.squarespace.com/static/5d2c7fcb94060e0001e56248/t/627cd5c8eab77d1e1693fab1/1652348403875/VCS_Project+Description_1201_10Jan2021_submitted.pdf)



|   |                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.                                                                                                | <p>The activity method will be applied for the demonstration of additionality for each new Project Instance.</p> <p><b>Step 1: Regulatory Surplus</b></p> <p>The improved cookstoves distribution is not mandated by any government enforced law, policy, statute or other regulatory framework in the Country.</p> <p><b>Step 2: Positive List</b></p> <p>The project activity introduces efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves which will be distributed at zero cost to the households in new PI activities. The only project's source of revenue is the economic value of the carbon credits generated by the introduction of ICSs.</p> |
| 6 | Occur within one of the designated geographic areas specified in the project description.                                                                                                                                                                | The new PI activity will be located in the geographical area included in the project boundaries defined in the Section 1.13. The GPS coordinates of each beneficiary household will be noted and stored in a management system as evidence of PI implementation inside the project boundaries.                                                                                                                                                                                                                                                                                                                                                                                             |
| 7 | Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.                                                       | Every new PIs activity will be compliant with the eligibility criteria listed in this table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8 | Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body. | <p>Each new PI activity will be described in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria.</p> <p>The evidence of local stakeholders' engagement and baseline data surveys and collection inside the project areas are described in this Project Design Document ("PDD").</p> <p>The monitoring data will be stored in a data management system database that will allow the Validation and Verification Body ("VVB") to check the compliance with VCS Standard v4.7 and the methodology VM0050 v1.0.</p>                                              |

|    |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions). | The ownership of each new Project activity Instances will be demonstrated by project proponent as the same described in Section 1.8                                                                                                                                                                                                                                                                                     |
| 10 | Have a start date that is the same as or later than the grouped project start date.                                                                                                                                                                                                  | The start date is the same than the grouped project start date, since the grouped project start date is the 1 <sup>st</sup> of June 2025.                                                                                                                                                                                                                                                                               |
| 11 | Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the total project crediting period                                       | <p>The project is eligible for crediting since the crediting period begins at the start of the project activity, when the first cookstove is distributed, as reported in section 1.9.</p> <p>The eligibility of possible additional PI will start from the later of either the start date of the project activity instance or the start of the verification period in which they were added to the grouped project.</p> |
| 12 | Not be or have been enrolled in another VCS project                                                                                                                                                                                                                                  | The grouped project and the correlated project activity instances are not enrolled in another VCS project, nor have either of them been enrolled in another VCS project.                                                                                                                                                                                                                                                |
| 13 | Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9                                                                                                                                                            | The PI adheres to the clustering requirements for multiple project activity instances. Moreover, the PI adhere to the capacity limit requirements for multiple project activity instances set out in the VCS Standard 4.7 (i.e. instances of the same project activity may not be spread across more than one project if they are within 10 Km of each other).                                                          |

## 1.6 Project Proponent

|                   |                                                |
|-------------------|------------------------------------------------|
| Organization name | Eni S.p.A.                                     |
| Contact person    | Lapo Niccolini                                 |
| Title             | Carbon credits portfolio management specialist |

|                  |                                                     |
|------------------|-----------------------------------------------------|
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|                          |                                                     |
|--------------------------|-----------------------------------------------------|
| <b>Organization name</b> | Eni S.p.A.                                          |
| <b>Contact person</b>    | Camilla Vuillermoz                                  |
| <b>Title</b>             | Carbon credits portfolio management specialist      |
| <b>Address</b>           | 20097 Via Emilia 1, San Donato Milanese (MI), Italy |
| <b>Telephone</b>         | +39 3758671566                                      |
| <b>Email</b>             | camilla.vuillermoz@eni.com                          |

## 1.7 Other Entities Involved in the Project

Eni has involved CUAMM, an Italian Non-Governmental Organization, for the field activities implementation. CUAMM supports Eni with community engagement, distribution of ICS and follow up and assistance to beneficiary households.

In the agreement signed between Eni and CUAMM, a clause regarding carbon credits rights has been included stating that Eni is the sole owner of carbon credits resulting from the project activities implementation and that CUAMM does not have any kind of claim on these.

|                            |                                            |
|----------------------------|--------------------------------------------|
| <b>Organization name</b>   | CUAMM (Doctors with Africa)                |
| <b>Role in the project</b> | Field implementing partner                 |
| <b>Contact person</b>      | Valerio Giuseppe                           |
| <b>Title</b>               | Field Project Manager                      |
| <b>Address</b>             | 22 Wilkinson Road – Freetown, Sierra Leone |
| <b>Telephone</b>           | +39 3924654942                             |
| <b>Email</b>               | g.valerio@cuamm.org                        |

## 1.8 Ownership

Eni S.p.A. is the owner of the project.

During the distribution of the ICSs to the households, a member for each household signs an EUA transferring to Eni the property rights of the carbon credits generated with the ICS.

Each ICS is labeled by a unique serial number attached to it that allows to identify every single device. Beneficiary households are informed during ICS training and sensibilization activities that they will receive the ICS free of any cash payment as part of this clean cooking project that Eni is developing in the Country with the goal of generating carbon credits through the distribution and use of improved cooking techniques.

The EUA signed by the recipient households contains the personal information of the beneficiary (i.e., birth date, address, ID document etc.), the house's GPS coordinates, and the serial number of the ICS provided. Moreover, Eni's carbon credits ownership is clearly stated in the **EUA** with the following wording:

*"The recipient accepts the assignment by Eni S.p.A. ("Eni") of the improved cookstove identified below ("cookstove") without any up-front cash payment undertaking to follow the conditions in this contract....[]*

*[]...The recipient waives any rights on the environmental benefit in favour of Eni: the Recipient declares that he/she has understood the mechanism for the carbon credits generation and waives any claim concerning the ownership of the carbon credits that will be generated using the cookstove".*

## 1.9 Project Start Date

|                    |                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Project start date | 1 <sup>st</sup> June 2025                                                                                                                |
| Justification      | The GP started in June 2025 when the first ICS of PI1, is delivered. Effective start date will be update following the validation phase. |

## 1.10 Project Crediting Period

|                                                       |                                                                                                                                             |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Crediting period                                      | Ten years, fixed                                                                                                                            |
| Start and end date of first or fixed crediting period | 1 <sup>st</sup> June 2025 to 31 <sup>st</sup> May 2035<br><br>The effective crediting period will be update following the validation phase. |

## 1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The GHG Emissions Reductions (ERs) have been computed assuming an initial number of ICS planned for the PI1 equal to 1.641.138, considering the ICS' parameters declared by local stoves manufacturer, such as the thermal efficiency and the expected lifespan, which have been certified by a CCA-recognized laboratory<sup>3</sup>.

- **Centre for Study and Research in Renewable Energy (CERER)** - University Cheikh Anta DIOP of Dakar

Regarding the actual operation of ICSs for the ERs ex-ante calculation: the ICS that reaches the expected lifetime (5-year lifetime) is considered as *non-operating*. In addition, considering the possible loss of operational devices even before the end of their lifespan (i.e. ICS broken or damaged, beneficiary household not using the ICS to its potential etc.), a decreasing ICSs operation rate ("drop off rate") of 5% per year has been assumed and included in the ER calculation for the 10-year crediting period.

During the monitoring activities for the ex-post emissions reduction calculation, the ICSs still operating after their expected lifespan may be considered for crediting during the project lifetime; analogously ICSs found out of service before the end of their lifespan will be discounted. The drop off rate parameter will be update at every monitoring period.

The estimated GHG emission reduction expected at entire project lifetime is almost 1,641 MtCO<sub>2</sub>e with an annual average GHG emission reduction lower than 300,000 tCO<sub>2</sub>e/year (project).

| Calendar year of crediting period   | Estimated GHG emission reductions or removals (tCO <sub>2</sub> e) |
|-------------------------------------|--------------------------------------------------------------------|
| 01-June-2025 to 31-December-2025    | 7.801,52                                                           |
| 01-January-2026 to 31-December-2026 | 47.924,26                                                          |
| 01-January-2027 to 31-December-2027 | 102.173,33                                                         |
| 01-January-2028 to 31-December-2028 | 160.656,05                                                         |
| 01-January-2029 to 31-December-2029 | 230.269,38                                                         |
| 01-January-2030 to 31-December-2030 | 302.362,77                                                         |
| 01-January-2031 to 31-December-2031 | 291.943,19                                                         |
| 01-January-2032 to 31-December-2032 | 223.969,99                                                         |
| 01-January-2033 to 31-December-2033 | 157.379,36                                                         |

<sup>3</sup> <https://cleancooking.org/regional-testing-and-knowledge-centers/>

|                                                                 |              |
|-----------------------------------------------------------------|--------------|
| 01-January-2034 to 31-December-2034                             | 90.159,54    |
| 01-January-2035 to 31-May-2035                                  | 26.499,10    |
| Total estimated ERRs during the first or fixed crediting period | 1.641.138,48 |
| Total number of years                                           | 10           |
| Average annual ERRs                                             | 164.113,85   |

## 1.12 Description of the Project Activity

The project aims to distribute improved cookstoves to households that are still relying on inefficient cooking techniques (i.e., three-stone fire or rudimental stove).

The ICSs will reduce GHG emissions from woody biomass combustion thanks to their higher thermal efficiency, with a direct reduction in firewood consumption. The ICS design allows to increase the heat transfer to cooking pots by providing a higher thermal insulation of stove's materials and a forced air flow conveyance through the stove mouth.

The switch of cooking systems leads to many co-benefits, first a decrease in woody biomass consumption reducing the pressure on forest cutting in the project area. The reduction of woody biomass necessary for cooking will result in time savings for women and children who are usually involved in biomass collection, allowing them to spend time for other activities.

Moreover, households living in villages far from forests will also save money since they will have to buy smaller quantities of wood for cooking.

Furthermore, considering that sometimes the households cook inside their homes, the substitution of traditional cooking mechanisms with ICSs could significantly reduce indoor air pollution with a direct positive impact on the health of household components.

The local communities in the project area are engaged on the framework activities starting from the stakeholder consultations, that gave the population the opportunity to information regarding the project implementation and ask questions/clarifications.

The households receive the ICS free of cash payment. A member of each household involved signs an EAU with Eni waiving the environmental benefits resulting from the use of the device back to Eni.

Every household in the Project Area would be involved in the ICSs distribution if it meets the following requirements:

- current use of woody biomass (firewood or charcoal) as cooking fuel;
- lack of access to cleaner cooking fuels (i.e. gas or electricity);

- lack of efficient cooking device (i.e. intermediate cookstove);
- non-participation in other ICS distribution programs.

The Project Proponents collects the personal data of the designated household's member, taking notes of the house's GPS location and the unique ID number of the delivered ICS. According to the methodology, those data are building up a database of all the beneficiaries which will be the basis for the monitoring and verification activities.

When the household receives the ICS, a dedicated training session is conducted to explain to the family members how the device works and how it is to be used and maintained.

Considering the current cooking fuel utilization (firewood), each household involved in the project will receive an ICS powered by the same biomass they currently use as fuel. The cookstoves will be locally produced. The actual production capacity of the company will gradually increase to assure that the production meets the targets of the project in terms of ICSs volume. Eventually, CUAMM will manage the distribution of the ICSs produced, including the storage of the cookstoves prior to distribution. The cookstove model distributed is shown in Figure 1.



Figure 1: Domestic Wood Burning cookstove produced in Sierra Leone

The ICS has an average lifetime of 5 years and a warranty period of 2 years. There will be multiple ICSs deliveries to cover the overall project lifespan (10 years).

- Independent tests on the efficiency of the ICS will be performed by “Centre for Study and Research in Renewable Energy (CERER) - University Cheikh Anta DIOP of Dakar” according to the Water Boiling Test protocol (WBT: ISO 19867:2018). The outcomes of the test said that the thermal efficiency is 36,9% for the firewood ICS, with a cooking power of 1,4kW and a fuel burning rate of 15,2 g/min.

In the following table the main technical features of the cookstove model produced are presented.

| Characteristics        |       |
|------------------------|-------|
| Thermal efficiency (%) | 36,9% |
| Lifetime (y)           | 5     |

Table 3: Technical specifications of the ICS distributed in the project (manufacturer specifications)

According to the baseline scenario, non-renewable woody biomass (wood) is being used by households for their cooking activities with very inefficient appliances (three-stone fire or traditional bracers). Without this clean cooking project implementation, the woody biomass consumption trend would increase with consequent harmful effects on deforestation and loss of biodiversity in the Country.

### 1.13 Project Location

The project is deployed in the Southern region of Sierra Leone, in the districts of Pujehun, Kenema and Kaliahun.

Pujehun district geographical coordinates are the following:

- 7°22'45"N
- 11°37'06"W

Kenema district geographical coordinates are the following:

- 7°54'24"N
- 11°15'04"W

Kaliahun district geographical coordinates are the following:

- 8°04'52"N
- 10°42'36"W





Figure 2: Geographical map of Sierra Leone with regions representation

## 1.14 Conditions Prior to Project Initiation

The conditions prior to the GP start are equal to the baseline scenario in the overall project area: utilization of inefficient rudimental braziers or three-stone fire burning non-renewable woody biomass (firewood). Please refer to Section **Errore. L'origine riferimento non è stata trovata.** for more details.

## 1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

In the Country there are not mandatory laws, policies or regulations which would oblige households to use energy efficient cookstoves. In the Country's climate mitigation strategy, elaborated before COP21 to the UNFCCC, a national clean cooking strategy and action plan was developed in 2020. However, these issues are not subject to binding targets.<sup>4</sup>

## 1.16 Double Counting and Participation under Other GHG Programs

<sup>4</sup> <https://faolex.fao.org/docs/pdf/sie212615.pdf>

### 1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

### 1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

### 1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

## 1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

### 1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

### 1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

### 1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

## 1.18 Sustainable Development Contributions

The distribution of ICSs is one of the activities that may accelerate the access to “clean cooking”, an urgent development issue with significant benefits for public health, gender equality,

deforestation tackling and climate change fight. The specific activities implemented in the project that will lead to Sustainable Development (“SD”) contributions are:

- Reduced use of biomass fuel for cooking activities;
- Reduced time spent for cooking and collecting fuel;
- Reduced smoke and GHG emissions from cooking activities;
- Engagement and training process of local communities on health, safety and environmental issues.

All these outputs are achieved by promoting and facilitating the switch to more efficient cooking technologies. The GP will lead to the following sustainable development contributions, as stated below:

- **SDG3: Ensure healthy lives and promote well-being for all at all ages.**

According to UNDP, the use of polluting fuels and technologies in traditional cooking systems is causing respiratory illness, heart problems and even deaths in many developing Countries. The life expectation in Sierra Leone is 60 years<sup>5</sup>. 52% of deaths from stroke and ischemic heart disease are caused by air pollution<sup>6</sup>, resulting in 7,334<sup>7</sup> deaths annually.

- **SDG5: Achieve gender equality and empower all women and girls.**

The energy sector in Sud-Saharan countries has been widely studied from a gender perspective. The women have the responsibility for domestic tasks, including fuel provisioning and cooking. The use of improved cookstoves would reduce the current practices that are very time consuming. The possibility to save time can engage women in other income generating activities, pursue educational opportunities or simply rest and spend time with family. Moreover, the impact is also related to the women exposure to assault during the collection trips.

- **SDG7: Ensure access to affordable, reliable, sustainable and modern energy for all.**

Clean cooking activities are crucial to achieve SDG7. According to the World Health Organization, in 2023, the population’ share without clean fuels and technology for cooking in Sierra Leone was 99%<sup>8</sup>. In the report “State of Access to Modern Energy

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<sup>5</sup> <https://data.worldbank.org/indicator/SP.DYN.LE00.IN?locations=SL>

<sup>6</sup> <https://cdn.who.int/media/docs/default-source/country-profiles/environmental-health/environmental-health-sle-2023.pdf#:~:text=1%20Air%20pollution%20causes%20many%20other>

<sup>7</sup> <https://www.who.int/data/gho/data/indicators/indicator-details/GHO/household-air-pollution-attributable-deaths>

<sup>8</sup> <https://cdn.who.int/media/docs/default-source/country-profiles/environmental-health/environmental-health-sle-2023.pdf#:~:text=1%20Air%20pollution%20causes%20many%20other>

Cooking Services” (World Bank, 2020)<sup>9</sup>, the main obstacles are a combination of underdeveloped infrastructures, high up-front capital costs and low households’ awareness. In the past decades, the interventions to increase energy access have not been sufficient to keep the path with population growth: therefore, the number of households accessing electricity has grown, but also the absolute number of people cooking with traditional stoves that burn biomass inefficiently has increased. The distribution of ICSs with a thermal efficiency twice as high as traditional cooking appliances allows local communities to reduce the biomass collected/purchased, with a direct impact on the households’ economic budget.

- **SDG13: Take urgent action to combat climate change and its impact.**

The less polluting cooking technology and the decreasing rate in deforestation are both actions to deal with climate change impacts. In the Country’s Intended Nationally Determined Contributions submitted in 2021, Sierra Leone declared its intention to reduce its GHG emissions by around 5% by 2025 and by 10% by 2030 relative to a business-as-usual trajectory<sup>10</sup>.

## 2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

## 3 APPLICATION OF METHODOLOGY

### 3.1 Title and Reference of Methodology

| Type (methodology, tool or module). | Reference ID, if applicable | Title                                                    | Version |
|-------------------------------------|-----------------------------|----------------------------------------------------------|---------|
| Methodology                         | VM0050                      | Energy efficiency and fuel-switch measures in cookstoves | 1.0     |

<sup>9</sup> [https://documents1.worldbank.org/curated/en/937141600195758792/pdf/The-State-of-Access-to-Modern-Energy-Cooking-Services.pdf?\\_gl=1\\*15cgo4m\\*\\_gcl\\_au\\*MTAyNTI5NDM4NS4xNzI2NzU0NzE2](https://documents1.worldbank.org/curated/en/937141600195758792/pdf/The-State-of-Access-to-Modern-Energy-Cooking-Services.pdf?_gl=1*15cgo4m*_gcl_au*MTAyNTI5NDM4NS4xNzI2NzU0NzE2)

<sup>10</sup> <https://www.iea.org/policies/18656-revisedupdated-ndc-of-sierra-leone>

|      |             |                                                   |     |
|------|-------------|---------------------------------------------------|-----|
| Tool | CDM TOOL 30 | Calculation of the fraction non-renewable biomass | 4.0 |
| Tool | CDM TOOL 33 | Default values for common parameters              | 2.0 |

## 3.2 Applicability of Methodology

| Methodology ID | Applicability condition                                                                                                                                                        | Justification of compliance                                                                                                                                                                                                                                                                                                |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VM0050         | <i>The project activity corresponds to replacement of non-renewable biomass fired cookstoves with more efficient project devices that use the same fuel as in the baseline</i> | <i>GP sees the distribution of improved cookstoves to households that are currently relying on inefficient cooking techniques (i.e. three-stone fire or traditional braziers). The households maintain the same woody biomass as cooking fuel (wood), reducing the necessary amount for thermal efficiency improvement</i> |
| VM0050         | <i>Methodology applicable to both 'Projects' and 'Large Projects'</i>                                                                                                          | <p><i>The GP size matches the "Project" definition criteria according to the rules set out in "VCS Standard v4.7".</i></p> <p><i>The estimated annual emission reduction for the PI1 is on average around 164.113 tCO<sub>2</sub>e per year.</i></p>                                                                       |
| VM0050         | <i>Project devices are used in households, communities, institutions, or small or medium enterprises (SMEs)</i>                                                                | <i>The improved cookstoves included in the GP activities will be distributed to households for domestic and/or small enterprise activities</i>                                                                                                                                                                             |
| VM0050         | <i>Project devices using non-renewable biomass, single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%</i>                   | <p><i>The ICSs are purchased from local manufacturers which produce among the others single pot stoves powered by firewood.</i></p> <p><i>The thermal efficiency of these ICSs, according to the</i></p>                                                                                                                   |

|        |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                                                                                                                                                                                                                                                                                                                   | <i>manufacturer's specifications and certified test results, is higher than 25%.</i>                                                                                                                                                                                                                                                                                                                                                                                 |
| VM0050 | <i>Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements, otherwise no further emission reductions may be claimed for the project device-</i> | <i>The delivered ICSs come with a warranty period of two years. Additionally, given the lifespan of the devices, they will no longer be included in the emission reduction calculations after five years.</i>                                                                                                                                                                                                                                                        |
| VM0050 | <i>The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description</i>                                                                                                   | <i>The mechanisms are implemented by the project partner and includes trainings sessions for lead mothers. The training covers financial management, the use and maintenance of improved cooking stoves, and lessons on health, nutrition, and environmental awareness. This approach empowers lead mothers to educate their communities on these topics that lead to behavioural changes.</i>                                                                       |
| VM0050 | <i>Explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).</i>                                                                                               | <i>To avoid double counting of emissions reduction resulting from cookstoves distributed within the GP boundaries, each ICS is identified and labelled with a unique serial number allowing for a clear distinction between the devices and ensuring that ICS are counted only once in the monitoring of the emission reductions. The serial number applied on the cookstove is stated in the End User Agreement signed by a designated member for the recipient</i> |

|        |                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                                                                                                                                                                                                                                                                                                                                                                  | <p>household, together with all other necessary data (i.e. ID number, telephone number, geographical coordinates of their house) establishing a unique association between the household and the ICS distributed.</p> <p>All those data will be collected and stored in a data management system to facilitate their control and check.</p>                                                                             |
| VM0050 | National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented.                                                                                                                                                                                                                | There is no national specific regulatory framework in place.                                                                                                                                                                                                                                                                                                                                                            |
| VM0050 | Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. | There is currently only one Carbon Offset REDD project underway in the Country: Gola REDD+. This project is located in the southeastern districts of Kailahun, Pujehun and Kenema, where households benefiting from the improved cookstoves do not rely on cutting woody biomass. Moreover, the areas covered by the Gola REDD+ project are remote and sparsely populated, so cookstoves will not be distributed there. |